Suraj Kumar Kothuri

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Google Scholar: scholar.google.com/citations?user=dAsh_q0AAAAJ

PROFILE

Biomedical optics engineer (PhD in Biophotonics, graduating Jan 2027) with four years of experience designing, building and validating compact optical systems for health monitoring. Built a low-cost, wearable time-domain near-infrared spectroscopy (TD-NIRS) tissue oximetry system from LiDAR components, taking it from concept to prototype to in vivo validation. Expertise: diffuse optics, tissue optics, light–tissue interaction, free-space optical design, single-photon detection, miniaturization, multimodal system integration, system validation on tissue phantoms.

CORE SKILLS

- **Optical systems:** Free-space optical design, optical alignment, lasers, single-photon detectors, silicon photomultipliers (SiPM), time-correlated single-photon counting (TCSPC), diffuse optics, time-domain near-infrared spectroscopy (TD-NIRS)
- **Biomedical optics:** Tissue optics, light–tissue interaction, tissue optical properties, time of flight spectroscopy, tissue oximetry, wearable health sensing
- **Design and simulation:** Zemax, SolidWorks (3D CAD), COMSOL Multiphysics, OptiFDTD
- **Cleanroom and photonics packaging:** Fiber array alignment, wire bonding, flip-chip bonding
- **Prototyping and integration:** Miniaturization, multimodal system integration, 3D printing
- **Test and validation:** System verification and validation, benchmarking on tissue phantoms, in vivo human studies, ex vivo studies
- **Human studies and research ethics:** Ethics applications, study protocols and consent documentation, research data management, ethics committee liaison
- **Software:** Python, MATLAB for data analysis, instrument automation and system validation

PROFESSIONAL EXPERIENCE

PhD Researcher – Biophotonics | Tyndall National Institute, Cork, Ireland **Feb 2023 – Present**

Wearable time-domain tissue oximetry

- Designed and built a time-domain near-infrared spectroscopy (TD-NIRS) system for tissue oxygenation monitoring, taking it from concept through prototype to system validation.
- Built the system from low-cost LiDAR components, making it cheaper than conventional TD-NIRS instruments and suitable for wearable biomedical applications.
- Demonstrated proof of concept in an in vivo study, measuring arm-muscle oxygenation changes during and after cuff occlusion.

Multimodal endoscopic probe

- Integrated a time-of-flight optical channel into a 4.0 mm side-viewing endoscopic probe alongside a chip-on-tip micro-camera.
- Benchmarked probe stability, repeatability and robustness on tissue phantoms and in vivo in human volunteers; coordinated phantom design with industry partners.

Optical characterization of human bone

- Led the first broadband (500–1100 nm) time-domain diffuse optical characterization of fresh human cadaver bone (published in nature Scientific Reports, 2025).

Research ethics and human studies

- Prepared and secured research ethics approval for light-based studies on human volunteers, liaising with Cork Research Ethics Committee.
- Planned and coordinated volunteer studies from ethics approval through data collection, including research data management.

Leadership and collaboration

- Mentored graduate and master's students through their final-year projects.
- Served as PhD student representative on the Center for Doctoral Training (CDT) management board.

- Collaborated with cross-functional teams across optics, spectroscopy, data analysis and system integration, managing multiple research projects in parallel.

EDUCATION

PhD in Biophotonics | Tyndall National Institute, University College Cork **Feb 2023 – Expected Jan 2027**
Thesis: Photonic Integration of Time-of-Flight Tools for Next-Generation Wearable Devices for Biomedical Applications

Integrated Masters in Physics (Photonics) | Sri Sathya Sai Institute of Higher Learning, India **2016 – 2022**

PROFESSIONAL TRAINING

- **Introduction to Biophotonics** – University College Cork
- **Photonics Packaging** – Tyndall National Institute, 3-day course, theory and hands-on lab
- **Compound Semiconductor Device Fabrication** – Tyndall National Institute, 5-day course, theory and hands-on lab
- **Micro-Transfer Printing** – X-Celeprint and Tyndall National Institute, 2-day course

SELECTED PUBLICATIONS

- "Multimodal endoscopy platform combining superficial tissue chip-on-tip micro-camera imaging and depth-resolved time-domain diffuse optical spectroscopy." *Journal of Physics: Photonics* (2026).
- "Broadband time domain diffuse optical characterization of human cadaver bone from 500 to 1100 nm." *Scientific Reports* (Nature Portfolio) (2025).

INTERESTS

Audio recording engineering, books, tech and AI